Tarea3_Arevalo_Arancibia

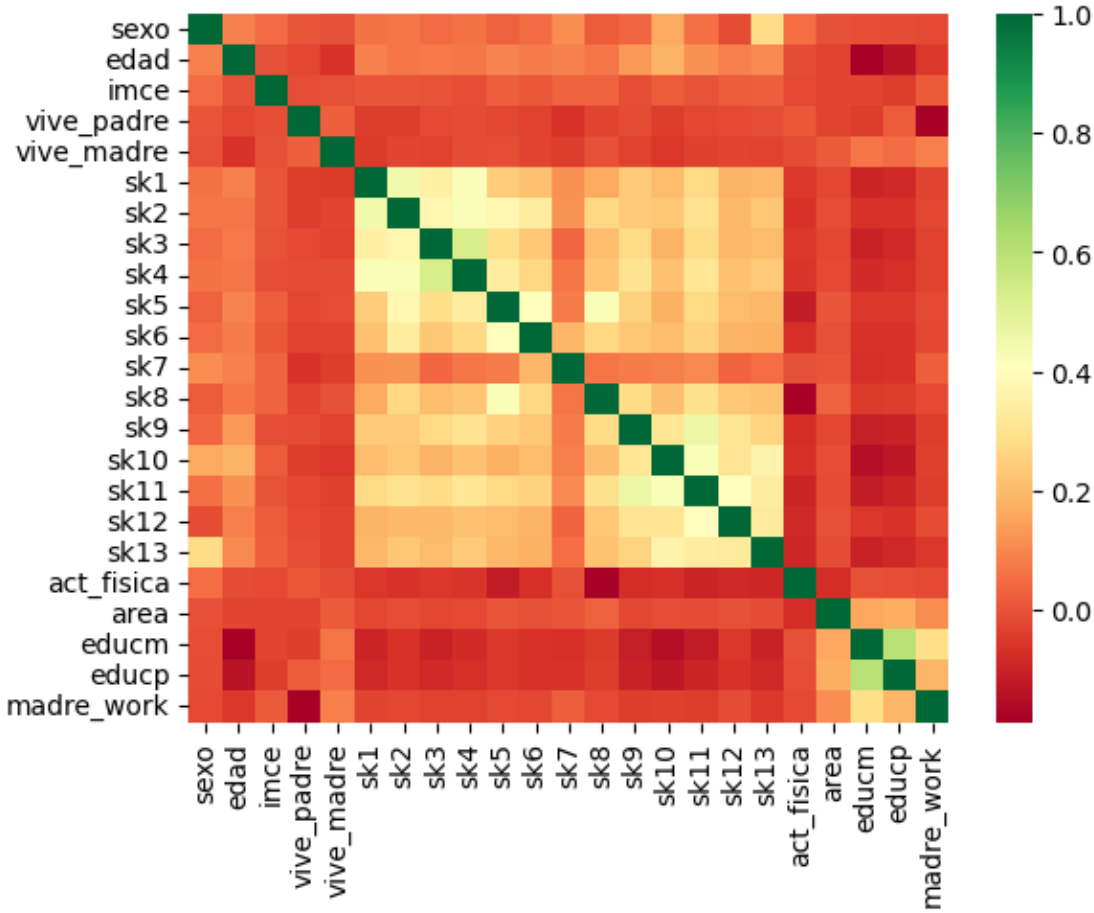
July 5, 2025

1 Tarea 3

1.1 1. Cargue la base de datos y realice los ajustes necesarios para su uso (missing values, recodificar variables, etcetera). Identifique los tipos de datos que se encuentran en la base, realice estadisticas descriptivas sobre las variables importantes (Hint: Revisar la distribuciones, datos faltantes, outliers, etc.) y limpie las variables cuando sea necesario.

```
               sexo          edad          imce    vive_padre    vive_madre  \
count  31055.000000  31055.000000  31055.000000  31055.000000  31055.000000
mean       0.538979     82.589084      1.017787      0.720335      0.966511
std        0.498486      4.360270      1.370552      0.449415      0.189671
min        0.000000     62.000000     -5.020000      0.000000      0.000000
25%        0.000000     80.000000      0.120000      0.000000      1.000000
50%        1.000000     81.000000      0.970000      1.000000      1.000000
75%        1.000000     82.000000      1.930000      1.000000      1.000000
max        1.000000    107.000000      5.040000      2.000000      2.000000

               sk1           sk2           sk3           sk4           sk5  \
count  31055.000000  31055.000000  31055.000000  31055.000000  31055.000000
mean       1.114088      1.391177      1.262856      1.256352      1.268588
std        0.394004      0.650190      0.585766      0.578266      0.564929
min        1.000000      1.000000      1.000000      1.000000      1.000000
25%        1.000000      1.000000      1.000000      1.000000      1.000000
50%        1.000000      1.000000      1.000000      1.000000      1.000000
75%        1.000000      2.000000      1.000000      1.000000      1.000000
max        5.000000      5.000000      5.000000      5.000000      5.000000

       …           sk9          sk10          sk11          sk12  \
count  …  31055.000000  31055.000000  31055.000000  31055.000000
mean   …      1.333666      1.853743      1.385349      1.502785
std    …      0.666009      0.941171      0.668783      0.799375
min    …      1.000000      1.000000      1.000000      1.000000
25%    …      1.000000      1.000000      1.000000      1.000000
50%    …      1.000000      2.000000      1.000000      1.000000
75%    …      1.000000      2.000000      2.000000      2.000000
max    …      5.000000      5.000000      5.000000      5.000000
```

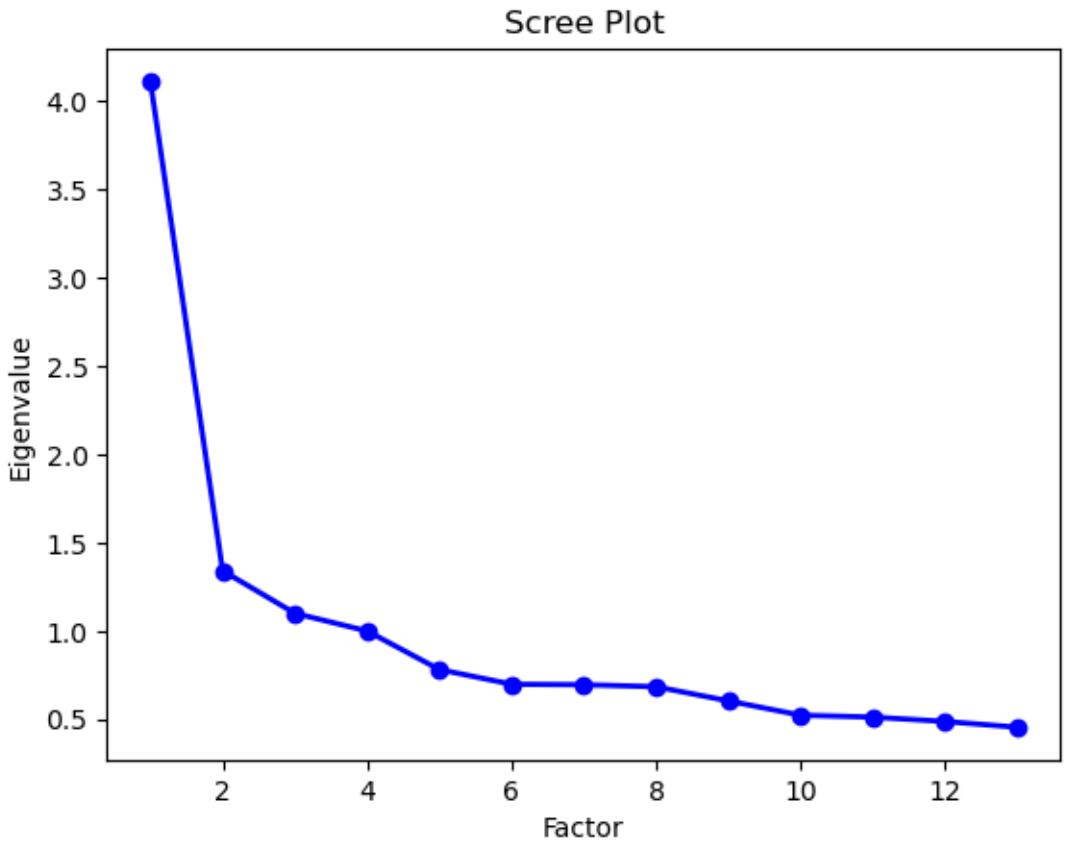
```
               sk13    act_fisica          area         educm         educp  \
count  31055.000000  30051.000000  31055.000000  30682.000000  31055.000000
mean       1.694735      2.556620      0.911544     13.069291     12.955756
std        0.984946      1.068041      0.283961      3.338320      3.425190
min        1.000000      1.000000      0.000000      0.000000      0.000000
25%        1.000000      2.000000      1.000000     12.000000     11.000000
50%        1.000000      2.000000      1.000000     13.000000     13.000000
75%        2.000000      3.000000      1.000000     15.000000     14.000000
max        5.000000      5.000000      1.000000     22.000000     22.000000

         madre_work
count  31055.000000
mean       0.092803
std        0.943509
min       -1.000000
25%       -1.000000
50%        0.000000
75%        1.000000
max        1.000000

[8 rows x 23 columns]

<Axes: >
```



1.2 2. A partir de las variables sk1-sk13 realice un EFA. En particular determine el numero optimo de factores y reporte las variables que se asocian a cada factor (usando los factor loadings). Tambien discuta si existen variables que no son informativas. (Hint: para realizar un EFA, todas las variables deben estar representatadas en el mismo sentido logico. Si una carateristica es negativa debe ser invertida en la escala, de tal forma que todas las variables representen aspectos positivos).

	sk1	sk2	sk3	sk4	sk5	sk6	sk7	sk8	sk9	sk10	sk11	sk12	sk13
0	1	1	1	2	1	2	2	1	2	2	2	3	2
1	1	1	1	1	1	1	4	1	1	1	1	1	1
2	2	2	3	2	1	2	3	2	2	3	2	1	3
3	1	1	1	1	1	1	2	1	1	1	1	1	1
4	1	1	1	1	1	1	3	1	1	1	1	1	1
…	…	…	…	…	…	…	…	…	…	…	…	…	…
31050	1	3	1	2	2	1	1	3	2	2	2	1	1
31051	1	1	2	2	1	1	1	2	1	3	2	1	4

```
31052    1    1    1    1    1    1    1    1    1     3     1     1     1
31053    1    1    1    1    2    2    1    2    1     2     1     1     2
31054    2    2    1    1    1    1    5    1    1     1     1     1     1

[31055 rows x 13 columns]

array([[0.183970, 0.549198, 0.155598],
       [0.177203, 0.517433, 0.350410],
       [0.182944, 0.596588, 0.168045],
       [0.198779, 0.686060, 0.200987],
       [0.121309, 0.219980, 0.724979],
       [0.173086, 0.210702, 0.481159],
       [0.069524, 0.074626, 0.136478],
       [0.269342, 0.104813, 0.478344],
       [0.495931, 0.215337, 0.207927],
       [0.563365, 0.122575, 0.127228],
       [0.664883, 0.209988, 0.205834],
       [0.526984, 0.116246, 0.144172],
       [0.485171, 0.154101, 0.134872]])

(array([4.114952, 1.341624, 1.100575, 1.000161, 0.781977, 0.698537,
        0.695787, 0.684453, 0.604042, 0.523209, 0.512081, 0.487796,
        0.454807]),
 array([3.526388, 0.743955, 0.517426, 0.211290, 0.114533, 0.062767,
        0.008348, -0.008400, -0.035529, -0.060437, -0.080399, -0.084919,
        -0.127254]))

array([0.359674, 0.421925, 0.417625, 0.550587, 0.588702, 0.305868,
       0.029029, 0.312344, 0.335551, 0.348592, 0.528531, 0.312011,
       0.277328])
```




```
(array([1.778781, 1.648043, 1.360946]),
 array([0.136829, 0.126773, 0.104688]),
 array([0.136829, 0.263602, 0.368290]))
```

```
eta1 =~ sk11 + sk9 + sk10 + sk12 + sk13 + sk4
eta2 =~ sk7 + sk6
eta3 =~ sk4 + sk11 + sk2 + sk5 + sk3 + sk9 + sk1 + sk8 + sk6 + sk13
```

```
FactorAnalyzer(rotation='varimax', rotation_kwargs={})
```

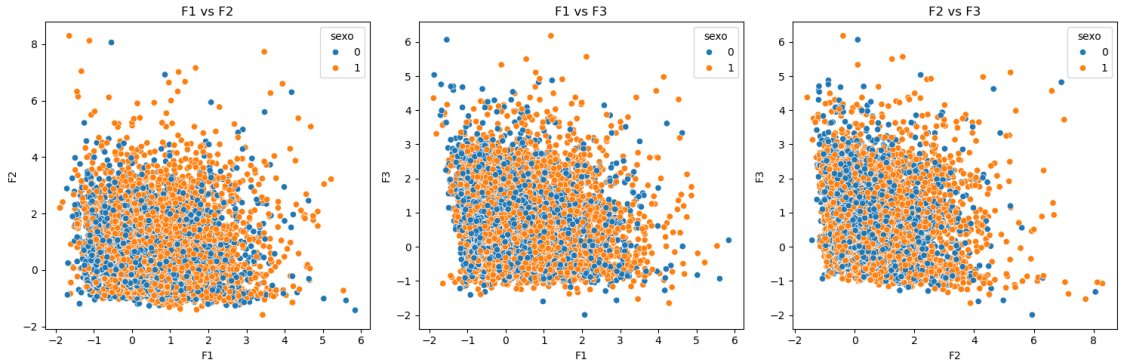
	Factor_1	Factor_2	Factor_3
sk1	0.183970	0.549198	0.155598
sk2	0.177203	0.517433	0.350410
sk3	0.182944	0.596588	0.168045
sk4	0.198779	0.686060	0.200987
sk5	0.121309	0.219980	0.724979
sk6	0.173086	0.210702	0.481159
sk7	0.069524	0.074626	0.136478
sk8	0.269342	0.104813	0.478344
sk9	0.495931	0.215337	0.207927

```
sk10   0.563365   0.122575   0.127228
sk11   0.664883   0.209988   0.205834
sk12   0.526984   0.116246   0.144172
sk13   0.485171   0.154101   0.134872
```

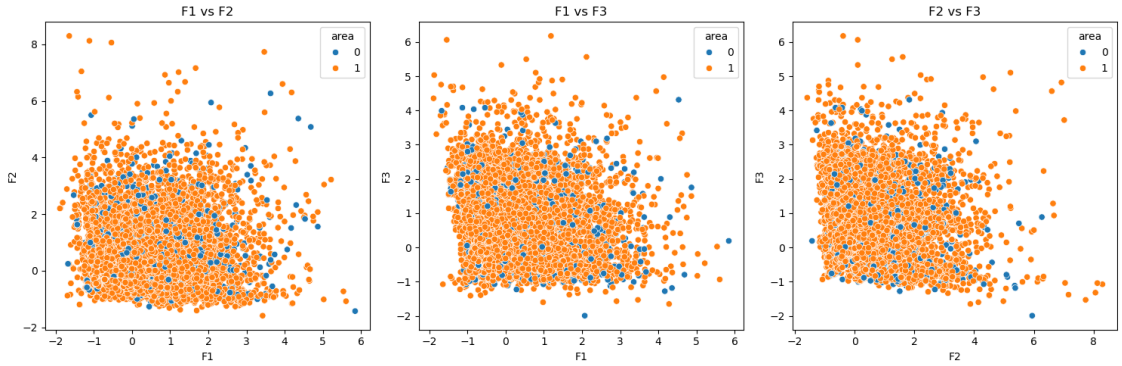
Respuesta: El nº de factores óptimos es 3 dado el Scree Plot. El factor no informativo dado el tamaño de la carga, es el sk7, por su baja carga en ambos de los 3 factores.

1.3 3. Con los resultados de la Pregunta 2, graficamente indique si existen diferencias significativas en la distribucion de los factores usando las siguientes variables para caracterizar: sexo, area, imce y act_fisica (Hint: usar scatterplot de factores en pares, y ocupar la opcion 'hue' para agregar variables de caracterizacion). Que puede concluir de los resultados?

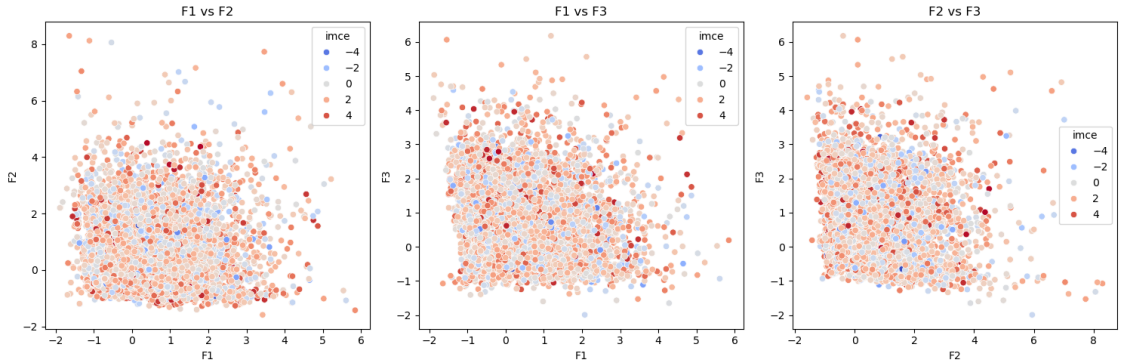
1.3.1 Por sexo



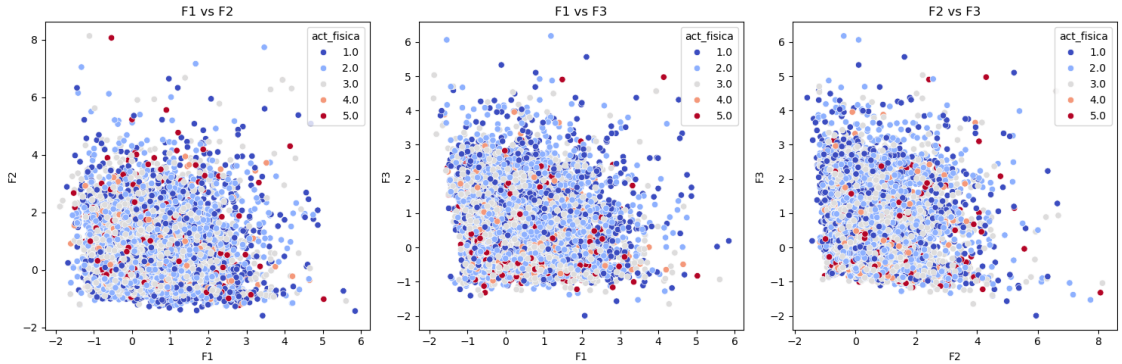
1.3.2 Por área



1.3.3 Por IMCE



1.3.4 Por Actividad Física



Respuesta: En ninguno de los indicadores y variables, se ve una diferencia significativa.

1.4 4. Con los resultados obtenidos en la Pregunta 2, proponga un CFA donde cada variable solo se asocia con un factor. Entregue un nombre a cada factor que representa el concepto comun entre las variables incluidas. Reporte la importancia de cada medida (variable) a cada factor e indique la correlacion entre factores (loadings).

```
Name of objective: MLW
Optimization method: SLSQP
Optimization successful.
Optimization terminated successfully
Objective value: 0.142
Number of iterations: 42
Params: 1.335 1.233 1.092 1.261 1.793 1.603 1.778 1.027 1.155 0.160 0.055 0.057
0.063 0.165 0.096 0.100 0.601 0.204 0.448 0.716 0.245 0.201 0.160 0.154 0.376
0.476 0.284
```

	F1	F2	F3
0	0.367608	0.052642	0.007638
1	-0.337317	-0.154623	-0.299660
2	0.467175	0.450303	0.284696
3	-0.337317	-0.154623	-0.299660
4	-0.337317	-0.154623	-0.299660
…	…	…	…
31050	0.243672	0.245844	0.425197
31051	0.299561	0.134912	0.017549
31052	-0.181009	-0.139361	-0.269122
31053	-0.117039	-0.079750	0.250202
31054	-0.277892	0.047391	-0.208849

[31055 rows x 3 columns]

	lval	op	rval	Estimate	Std. Err	z-value	p-value
0	sk9	~	F1	1.000000	-	-	-
1	sk10	~	F1	1.334713	0.017645	75.641962	0.0
2	sk11	~	F1	1.232807	0.013993	88.102986	0.0
3	sk12	~	F1	1.091950	0.014832	73.622089	0.0
4	sk13	~	F1	1.261417	0.017981	70.151539	0.0
5	sk1	~	F2	1.000000	-	-	-
6	sk2	~	F2	1.793157	0.021586	83.072223	0.0
7	sk3	~	F2	1.603422	0.019395	82.673817	0.0
8	sk4	~	F2	1.777960	0.020174	88.132298	0.0
9	sk5	~	F3	1.000000	-	-	-
10	sk6	~	F3	1.026632	0.013824	74.266085	0.0
11	sk8	~	F3	1.155310	0.015557	74.262593	0.0
12	F1	~~	F1	0.159913	0.003116	51.314941	0.0
13	F2	~~	F2	0.055129	0.001076	51.236155	0.0
14	F2	~~	F1	0.056757	0.001019	55.701576	0.0
15	F2	~~	F3	0.062711	0.00104	60.278918	0.0
16	F3	~~	F3	0.165310	0.002849	58.023053	0.0
17	F3	~~	F1	0.096470	0.001696	56.871328	0.0
18	sk1	~~	sk1	0.100118	0.000947	105.731758	0.0
19	sk10	~~	sk10	0.600837	0.005587	107.548234	0.0
20	sk11	~~	sk11	0.204229	0.002503	81.593079	0.0
21	sk12	~~	sk12	0.448268	0.0041	109.342606	0.0
22	sk13	~~	sk13	0.715675	0.006396	111.894751	0.0
23	sk2	~~	sk2	0.245293	0.002461	99.659864	0.0
24	sk3	~~	sk3	0.201387	0.002007	100.32833	0.0
25	sk4	~~	sk4	0.160143	0.001843	86.908808	0.0
26	sk5	~~	sk5	0.153880	0.002146	71.703273	0.0
27	sk6	~~	sk6	0.375940	0.003685	102.011723	0.0
28	sk8	~~	sk8	0.476185	0.004668	102.016687	0.0
29	sk9	~~	sk9	0.283549	0.002721	104.22048	0.0

F1 =~ sk9 + sk10 + sk11 + sk12 + sk13

```
F2 =~ sk1 + sk2 + sk3 + sk4
F3 =~ sk5 + sk6 + sk8
```

Respuesta: Todas las variables son significativas para el modelo, que usé filtrando los resultados de la ronda anterior. En el significado de los factores tenemos que:

- F1 representa su interés por conocer el mundo y expresarse a través de él.

- F2 representa su capacidad de gestionar emociones.

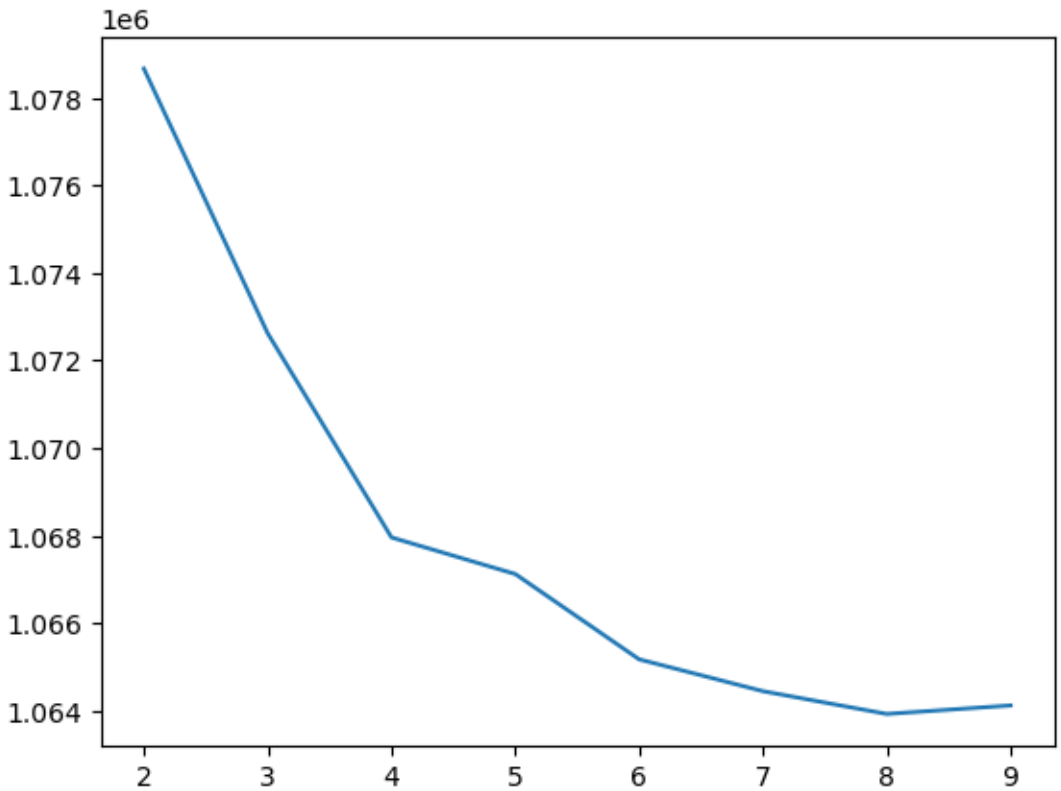
- F3 representa la capacidad de relacionarse con otros.

1.5 5. Estime un modelo de clases latentes usando imce via StepMix. Determine el numero optimo de clases (clusters) y muestre las diferencias entre clusters graficamente. Que se puede concluir de los resultados?

```
Fitting StepMix…

Initializations (n_init) :   0%|          | 0/1 [00:00<?, ?it/s]

Initializations (n_init) : 100%|     | 1/1 [00:05<00:00,  5.46s/it,
max_LL=-5.39e+5, max_avg_LL=-17.4]

Fitting StepMix…

Initializations (n_init) : 100%|     | 1/1 [00:09<00:00,  9.63s/it,
max_LL=-5.36e+5, max_avg_LL=-17.3]

Fitting StepMix…

Initializations (n_init) : 100%|     | 1/1 [00:18<00:00, 18.82s/it,
max_LL=-5.33e+5, max_avg_LL=-17.2]

Fitting StepMix…

Initializations (n_init) : 100%|     | 1/1 [01:33<00:00, 93.44s/it,
max_LL=-5.33e+5, max_avg_LL=-17.2]

Fitting StepMix…

Initializations (n_init) : 100%|     | 1/1 [00:57<00:00, 57.64s/it,
max_LL=-5.32e+5, max_avg_LL=-17.1]

Fitting StepMix…

Initializations (n_init) : 100%|     | 1/1 [01:16<00:00, 76.59s/it,
max_LL=-5.31e+5, max_avg_LL=-17.1]

Fitting StepMix…

Initializations (n_init) : 100%|     | 1/1 [01:40<00:00, 100.76s/it,
max_LL=-5.31e+5, max_avg_LL=-17.1]

Fitting StepMix…
```

```
Initializations (n_init) : 100%|      | 1/1 [02:43<00:00, 163.81s/it,
max_LL=-5.31e+5, max_avg_LL=-17.1]
/opt/anaconda3/lib/python3.12/site-packages/stepmix/stepmix.py:968:
ConvergenceWarning: Initializations did not converge. Try different init
parameters, or increase max_iter, abs_tol, rel_tol or check for degenerate data.
  warnings.warn(
```

```
<Axes: >
```

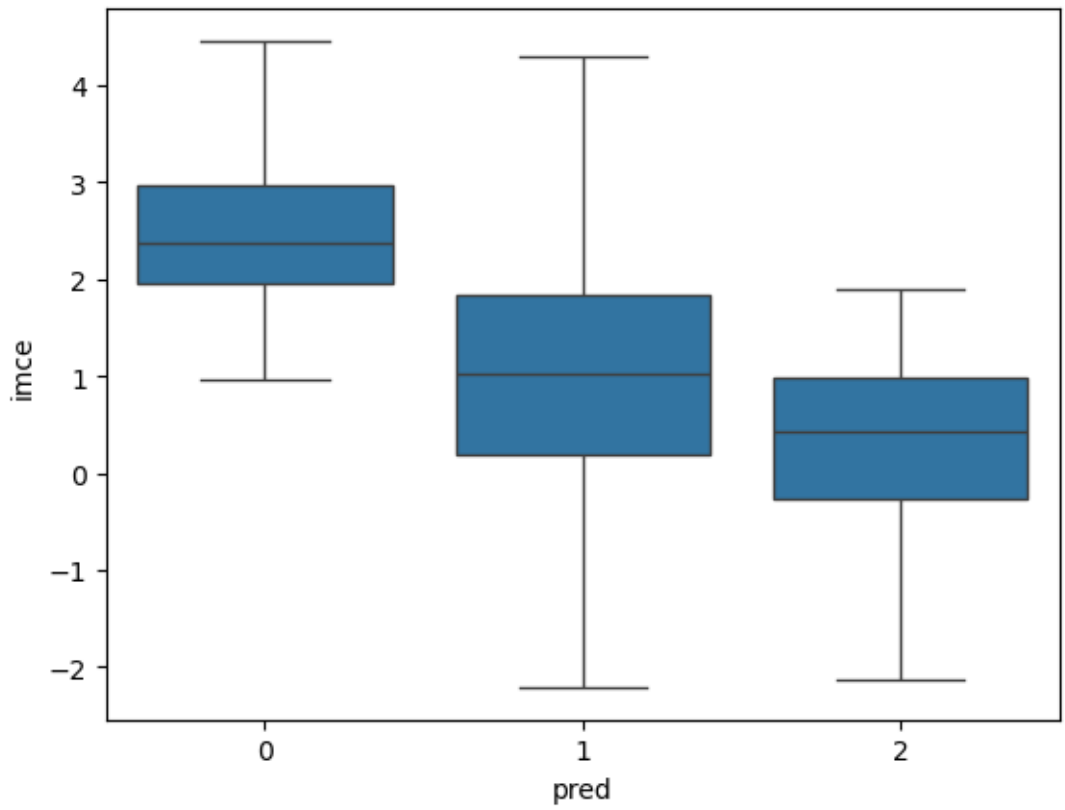



```
Fitting StepMix…

Initializations (n_init) :   0%|          | 0/1 [00:00<?, ?it/s]

Initializations (n_init) : 100%|      | 1/1 [00:10<00:00, 10.69s/it,
max_LL=-5.36e+5, max_avg_LL=-17.3]

================================================================================
MODEL REPORT
================================================================================
    ==============================================================================
    Measurement model parameters
    ==============================================================================
          model_name     gaussian_unit
```

```
      class_no                  0       1       2
      param variable
      means imce            2.1840  1.0301  0.3561
            sk1             1.0705  1.5259  1.0599
            sk10            1.7662  2.9693  1.6898
            sk11            1.2814  2.3588  1.2579
            sk12            1.4133  2.4156  1.3787
            sk13            1.5805  2.9613  1.5169
            sk2             1.3194  2.1236  1.2916
            sk3             1.1932  1.8818  1.1838
            sk4             1.1732  1.9548  1.1697
            sk5             1.2162  1.8956  1.1782
            sk6             1.4380  2.2205  1.3798
            sk7             2.3865  2.7165  2.0609
            sk8             1.5043  2.4062  1.4238
            sk9             1.2370  2.1828  1.2258
    ==============================================================================
    Class weights
    ==============================================================================
        Class 1 : 0.32
        Class 2 : 0.11
        Class 3 : 0.57
    ==============================================================================
    Fit for 3 latent classes
    ==============================================================================
    Estimation method             : 1-step
    Number of observations        : 31055
    Number of latent classes      : 3
    Number of estimated parameters: 44
    Log-likelihood (LL)           : -535926.8318
    -2LL                          : 1071853.6636
    Average LL                    : -17.2573
    AIC                           : 1071941.66
    BIC                           : 1072308.78
    CAIC                          : 1072352.78
    Sample-Size Adjusted BIC      : 1072624.06
    Entropy                       : 13065.6419
    Scaled Relative Entropy       : 0.6170
```

```
/var/folders/8q/08j5pdjn55n6s7dyy49m65w00000gn/T/ipykernel_1563/1065889366.py:6:
SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-
docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
  df_latente['pred']=model.predict(df_latente)
```

```
          imce
pred
0     2.528531
1     1.031688
2     0.279624

       sk1  sk2  sk3  sk4  sk5  sk6  sk7  sk8  sk9  sk10  sk11  sk12  sk13  \
0        1    1    1    2    1    2    2    1    2     2     2     3     2
1        1    1    1    1    1    1    4    1    1     1     1     1     1
2        2    2    3    2    1    2    3    2    2     3     2     1     3
3        1    1    1    1    1    1    2    1    1     1     1     1     1
4        1    1    1    1    1    1    3    1    1     1     1     1     1
…     …  …  …  …  …  …  …  …  …  …  …  …  …
31050    1    3    1    2    2    1    1    3    2     2     2     1     1
31051    1    1    2    2    1    1    1    2    1     3     2     1     4
31052    1    1    1    1    1    1    1    1    1     3     1     1     1
31053    1    1    1    1    2    2    1    2    1     2     1     1     2
31054    2    2    1    1    1    1    5    1    1     1     1     1     1

       imce  pred
0      0.75     2
1      0.71     2
2      0.27     1
3      2.05     0
4      1.05     2
…     …   …
31050  1.63     2
31051  2.57     0
31052  2.12     0
31053 -0.43     2
31054 -0.55     2

[31055 rows x 15 columns]

<Axes: xlabel='pred', ylabel='imce'>
```



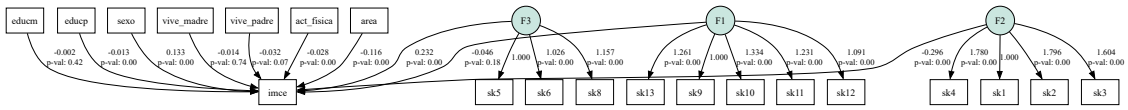
Respuesta: Dado los parametros se ven 3 grandes clústeres, el clúster "0" de media en el imce de sobrepeso, el clúster "1" de peso promedio y el clúster "2" de peso bajo la media.

1.6 6. Implemente un SEM completo usando la estructura propuesta en la Pregunta 4. En particular, estime un modelo donde los factores latentes explican el imce, junto con otras variables que existen en la base de datos. Ademas utilice otras variables relevantes de la base de datos para su modelo. Las variables a incluir en el modelo final deben tener sustento teorico y el modelo final debe optimizar el ajuste a los datos, en base a los criterios vistos en clase (TLI, CLI, RMSEA). Que puede concluir en base a sus resultados?

```
Name of objective: MLW
Optimization method: SLSQP
Optimization successful.
Optimization terminated successfully
Objective value: 0.341
Number of iterations: 52
Params: 1.334 1.231 1.091 1.261 1.796 1.604 1.780 1.026 1.157 -0.046 -0.296
0.232 0.133 -0.116 -0.028 -0.032 -0.014 -0.013 -0.002 0.160 0.055 0.057 0.063
```

0.165 0.097 1.865 0.100 0.601 0.204 0.448 0.716 0.245 0.201 0.160 0.154 0.376 0.476 0.283

	lval	op	rval	Estimate	Std. Err	z-value	p-value
0	sk9	~	F1	1.000000	-	-	-
1	sk10	~	F1	1.333870	0.017626	75.67699	0.0
2	sk11	~	F1	1.231354	0.01397	88.143665	0.0
3	sk12	~	F1	1.090943	0.014814	73.641994	0.0
4	sk13	~	F1	1.260510	0.017961	70.181981	0.0
5	sk1	~	F2	1.000000	-	-	-
6	sk2	~	F2	1.795516	0.021615	83.069766	0.0
7	sk3	~	F2	1.604009	0.01941	82.636552	0.0
8	sk4	~	F2	1.779715	0.020198	88.114357	0.0
9	sk5	~	F3	1.000000	-	-	-
10	sk6	~	F3	1.026101	0.01382	74.245736	0.0
11	sk8	~	F3	1.156548	0.015562	74.317906	0.0
12	imce	~	F1	-0.046491	0.03432	-1.35461	0.175542
13	imce	~	F2	-0.296183	0.065467	-4.524127	0.000006
14	imce	~	F3	0.232484	0.039767	5.846202	0.0
15	imce	~	sexo	0.132956	0.01558	8.533556	0.0
16	imce	~	area	-0.116369	0.027892	-4.17208	0.00003
17	imce	~	act_fisica	-0.027629	0.007292	-3.789185	0.000151
18	imce	~	vive_padre	-0.031700	0.017309	-1.831404	0.06704
19	imce	~	vive_madre	-0.013689	0.040978	-0.334044	0.738346
20	imce	~	educp	-0.013402	0.002842	-4.716386	0.000002
21	imce	~	educm	-0.002354	0.002912	-0.808517	0.418793
22	F1	~~	F1	0.160170	0.003119	51.353128	0.0
23	F2	~~	F2	0.055057	0.001075	51.206476	0.0
24	F2	~~	F1	0.056770	0.001019	55.707643	0.0
25	F2	~~	F3	0.062633	0.001039	60.256787	0.0
26	F3	~~	F3	0.165161	0.002847	58.019539	0.0
27	F3	~~	F1	0.096507	0.001697	56.88033	0.0
28	imce	~~	imce	1.865097	0.015017	124.199185	0.0
29	sk1	~~	sk1	0.100157	0.000947	105.774129	0.0
30	sk10	~~	sk10	0.600880	0.005587	107.545314	0.0
31	sk11	~~	sk11	0.204268	0.002502	81.636587	0.0
32	sk12	~~	sk12	0.448348	0.0041	109.35062	0.0
33	sk13	~~	sk13	0.715564	0.006395	111.891769	0.0
34	sk2	~~	sk2	0.245240	0.002461	99.632994	0.0
35	sk3	~~	sk3	0.201432	0.002007	100.360588	0.0
36	sk4	~~	sk4	0.160068	0.001842	86.884753	0.0
37	sk5	~~	sk5	0.153947	0.002144	71.794353	0.0
38	sk6	~~	sk6	0.376169	0.003685	102.088764	0.0
39	sk8	~~	sk8	0.475848	0.004666	101.986496	0.0
40	sk9	~~	sk9	0.283456	0.002721	104.182296	0.0



Respuesta: Agregando los factores antes calculados, queremos ver si el que los padres hayan sido educados, el sexo del niño, como es que la presencia de tanto padre como madre en sus vidas o el sector donde vivan influye en el físico del niño. Asi podemos ver que los 3 factores aportan al imce significativamente. Para las variables que agregué, la educación del padre, el sexo, vivir con el papá, la actividad física y el área son relevantes para el imce también. También es relevante ver que el sk5 no es relevante para el factor.

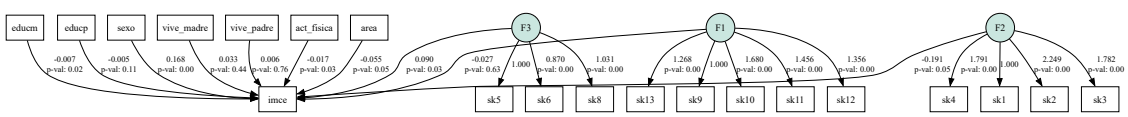
1.7 7. Repita el analisis de la Pregunta 6 usando para cada submuestra generada a partir de las clases latentes obtenida en la pregunta 5. Reporte diferencias entre los parametros entre los distintos grupos (si existieran). Hay alguna ganancia de estimar el modelo por grupos versus el modelo general?

1.7.1 Submuestra 1

```
Name of objective: MLW
Optimization method: SLSQP
Optimization successful.
Optimization terminated successfully
Objective value: 0.290
Number of iterations: 49
Params: 1.680 1.456 1.356 1.268 2.249 1.782 1.791 0.870 1.031 -0.027 -0.191
0.090 0.168 -0.055 -0.017 0.006 0.033 -0.005 -0.007 0.048 0.019 0.010 0.022
0.114 0.023 0.596 0.069 0.612 0.178 0.403 0.671 0.222 0.181 0.132 0.134 0.407
0.495 0.228
```

	lval	op	rval	Estimate	Std. Err	z-value	p-value
0	sk9	~	F1	1.000000	-	-	-
1	sk10	~	F1	1.679825	0.076289	22.019106	0.0
2	sk11	~	F1	1.456022	0.060703	23.986146	0.0
3	sk12	~	F1	1.356113	0.061705	21.977413	0.0
4	sk13	~	F1	1.268168	0.067388	18.819003	0.0
5	sk1	~	F2	1.000000	-	-	-
6	sk2	~	F2	2.248961	0.082058	27.407017	0.0
7	sk3	~	F2	1.781812	0.067275	26.48541	0.0
8	sk4	~	F2	1.791100	0.064926	27.586672	0.0
9	sk5	~	F3	1.000000	-	-	-
10	sk6	~	F3	0.869630	0.038419	22.635361	0.0
11	sk8	~	F3	1.031313	0.044656	23.094757	0.0
12	imce	~	F1	-0.026562	0.055888	-0.475273	0.634592
13	imce	~	F2	-0.190825	0.099117	-1.925249	0.054198
14	imce	~	F3	0.090207	0.040984	2.200997	0.027736
15	imce	~	sexo	0.167607	0.016409	10.214599	0.0
16	imce	~	area	-0.055017	0.02865	-1.92031	0.054819
17	imce	~	act_fisica	-0.017194	0.007861	-2.187229	0.028726
18	imce	~	vive_padre	0.005591	0.018	0.310602	0.756103

19	imce	~	vive_madre	0.033118	0.043166	0.767234	0.442943
20	imce	~	educp	-0.004684	0.002934	-1.59661	0.110353
21	imce	~	educm	-0.006887	0.003005	-2.292026	0.021904
22	F1	~~	F1	0.047539	0.003123	15.224143	0.0
23	F2	~~	F2	0.018756	0.001052	17.834248	0.0
24	F2	~~	F1	0.009504	0.000664	14.316974	0.0
25	F2	~~	F3	0.021518	0.00106	20.298112	0.0
26	F3	~~	F3	0.114461	0.005515	20.754585	0.0
27	F3	~~	F1	0.023112	0.001583	14.599003	0.0
28	imce	~~	imce	0.596025	0.00886	67.27427	0.0
29	sk1	~~	sk1	0.068704	0.001225	56.092049	0.0
30	sk10	~~	sk10	0.611665	0.010806	56.604853	0.0
31	sk11	~~	sk11	0.178051	0.004479	39.749951	0.0
32	sk12	~~	sk12	0.402611	0.007099	56.716315	0.0
33	sk13	~~	sk13	0.670929	0.010837	61.911319	0.0
34	sk2	~~	sk2	0.222479	0.004491	49.537349	0.0
35	sk3	~~	sk3	0.180684	0.003366	53.675643	0.0
36	sk4	~~	sk4	0.131798	0.00273	48.273983	0.0
37	sk5	~~	sk5	0.133897	0.004973	26.924264	0.0
38	sk6	~~	sk6	0.407224	0.007212	56.464938	0.0
39	sk8	~~	sk8	0.495254	0.009096	54.445771	0.0
40	sk9	~~	sk9	0.228368	0.003994	57.177497	0.0

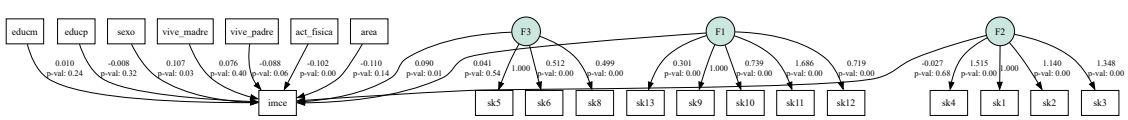


1.7.2 Submuestra 2

```
Name of objective: MLW
Optimization method: SLSQP
Optimization successful.
Optimization terminated successfully
Objective value: 0.440
Number of iterations: 37
Params: 0.739 1.686 0.719 0.301 1.140 1.348 1.515 0.512 0.499 0.041 -0.027 0.090
0.107 -0.110 -0.102 -0.088 0.076 -0.008 0.010 0.174 0.203 0.030 0.129 0.621
0.008 1.596 0.403 0.875 0.341 1.010 1.281 0.523 0.465 0.414 0.124 0.743 0.878
0.838
```

	lval	op	rval	Estimate	Std. Err	z-value	p-value
0	sk9	~	F1	1.000000	-	-	-
1	sk10	~	F1	0.738789	0.061326	12.04691	0.0
2	sk11	~	F1	1.685858	0.148734	11.334684	0.0
3	sk12	~	F1	0.719362	0.06361	11.308862	0.0
4	sk13	~	F1	0.301344	0.060307	4.996823	0.000001
5	sk1	~	F2	1.000000	-	-	-

6	sk2	~	F2	1.139987	0.048179	23.661592	0.0
7	sk3	~	F2	1.348289	0.052729	25.570168	0.0
8	sk4	~	F2	1.515260	0.057563	26.323741	0.0
9	sk5	~	F3	1.000000	-	-	-
10	sk6	~	F3	0.511578	0.037647	13.588927	0.0
11	sk8	~	F3	0.499246	0.037864	13.185265	0.0
12	imce	~	F1	0.040731	0.06667	0.610936	0.541242
13	imce	~	F2	-0.026762	0.064061	-0.417757	0.676125
14	imce	~	F3	0.089862	0.034323	2.618129	0.008841
15	imce	~	sexo	0.106841	0.04775	2.237481	0.025255
16	imce	~	area	-0.109689	0.073476	-1.492857	0.135474
17	imce	~	act_fisica	-0.101733	0.021053	-4.832167	0.000001
18	imce	~	vive_padre	-0.088476	0.047361	-1.868126	0.061744
19	imce	~	vive_madre	0.075871	0.089674	0.846076	0.397511
20	imce	~	educp	-0.007924	0.007911	-1.001674	0.316501
21	imce	~	educm	0.009534	0.008086	1.179019	0.238391
22	F1	~~	F1	0.174431	0.020569	8.480211	0.0
23	F2	~~	F2	0.202669	0.01297	15.626344	0.0
24	F2	~~	F1	0.030051	0.005264	5.708858	0.0
25	F2	~~	F3	0.129276	0.008877	14.563076	0.0
26	F3	~~	F3	0.620793	0.043754	14.188364	0.0
27	F3	~~	F1	0.008389	0.007744	1.08327	0.278689
28	imce	~~	imce	1.596276	0.039297	40.621119	0.0
29	sk1	~~	sk1	0.402908	0.011948	33.72067	0.0
30	sk10	~~	sk10	0.874596	0.023411	37.358839	0.0
31	sk11	~~	sk11	0.341153	0.041385	8.243317	0.0
32	sk12	~~	sk12	1.010414	0.026527	38.089881	0.0
33	sk13	~~	sk13	1.280936	0.031727	40.373215	0.0
34	sk2	~~	sk2	0.523445	0.015524	33.71844	0.0
35	sk3	~~	sk3	0.465176	0.015781	29.476352	0.0
36	sk4	~~	sk4	0.413802	0.016576	24.963224	0.0
37	sk5	~~	sk5	0.124052	0.039966	3.103937	0.00191
38	sk6	~~	sk6	0.743033	0.021199	35.051045	0.0
39	sk8	~~	sk8	0.878283	0.02397	36.640985	0.0
40	sk9	~~	sk9	0.837756	0.025545	32.795694	0.0

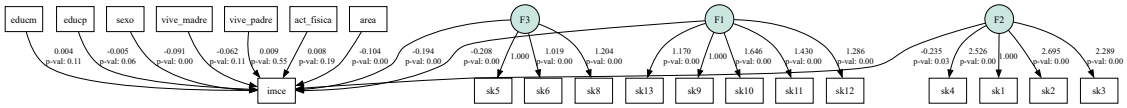


1.7.3 Submuestra 3

```
Name of objective: MLW
Optimization method: SLSQP
Optimization successful.
Optimization terminated successfully
```

```
Objective value: 0.276
Number of iterations: 55
Params: 1.646 1.430 1.286 1.170 2.695 2.289 2.526 1.019 1.204 -0.208 -0.235
-0.194 -0.091 -0.104 0.008 0.009 -0.062 -0.005 0.004 0.046 0.012 0.008 0.015
0.079 0.022 0.875 0.059 0.542 0.158 0.372 0.610 0.209 0.161 0.124 0.113 0.315
0.404 0.218
```

	lval	op	rval	Estimate	Std. Err	z-value	p-value
0	sk9	~	F1	1.000000	-	-	-
1	sk10	~	F1	1.645976	0.050645	32.500125	0.0
2	sk11	~	F1	1.429565	0.040524	35.277337	0.0
3	sk12	~	F1	1.286490	0.040499	31.765633	0.0
4	sk13	~	F1	1.170468	0.043865	26.683115	0.0
5	sk1	~	F2	1.000000	-	-	-
6	sk2	~	F2	2.694809	0.072181	37.334314	0.0
7	sk3	~	F2	2.288987	0.061801	37.038033	0.0
8	sk4	~	F2	2.525650	0.065711	38.435655	0.0
9	sk5	~	F3	1.000000	-	-	-
10	sk6	~	F3	1.018646	0.028563	35.663306	0.0
11	sk8	~	F3	1.204084	0.033365	36.088332	0.0
12	imce	~	F1	-0.208084	0.048922	-4.253348	0.000021
13	imce	~	F2	-0.234570	0.10722	-2.187743	0.028688
14	imce	~	F3	-0.193772	0.044325	-4.371618	0.000012
15	imce	~	sexo	-0.091042	0.013744	-6.62419	0.0
16	imce	~	area	-0.104232	0.025241	-4.129553	0.000036
17	imce	~	act_fisica	0.008387	0.006405	1.309517	0.190359
18	imce	~	vive_padre	0.009237	0.015461	0.597462	0.550199
19	imce	~	vive_madre	-0.062392	0.038538	-1.618982	0.105451
20	imce	~	educp	-0.004699	0.002545	-1.846527	0.064816
21	imce	~	educm	0.004193	0.002615	1.603702	0.10878
22	F1	~~	F1	0.046499	0.002088	22.266947	0.0
23	F2	~~	F2	0.011800	0.000514	22.969106	0.0
24	F2	~~	F1	0.007759	0.000371	20.9331	0.0
25	F2	~~	F3	0.015077	0.000526	28.652181	0.0
26	F3	~~	F3	0.078988	0.002582	30.588959	0.0
27	F3	~~	F1	0.021679	0.000949	22.85072	0.0
28	imce	~~	imce	0.874719	0.0091	96.127019	0.0
29	sk1	~~	sk1	0.058607	0.000685	85.589848	0.0
30	sk10	~~	sk10	0.542496	0.006715	80.78494	0.0
31	sk11	~~	sk11	0.157646	0.002813	56.037229	0.0
32	sk12	~~	sk12	0.372286	0.004507	82.597319	0.0
33	sk13	~~	sk13	0.609717	0.0068	89.664386	0.0
34	sk2	~~	sk2	0.208798	0.002836	73.624565	0.0
35	sk3	~~	sk3	0.161135	0.002143	75.178182	0.0
36	sk4	~~	sk4	0.124175	0.001979	62.752423	0.0
37	sk5	~~	sk5	0.112885	0.002335	48.337528	0.0
38	sk6	~~	sk6	0.315287	0.004032	78.190518	0.0
39	sk8	~~	sk8	0.404448	0.005296	76.370024	0.0
40	sk9	~~	sk9	0.218270	0.002657	82.150398	0.0



Respuesta: En materia de factores vemos que la educación es relevante solo en la submuestra 1, el sexo es relevante en las 3 submuestras, vivir con los padres no es relevante en ninguna, el area solo es relevante en la 3ra submuestra, mientras que para esta misma hacer actividad física no lo es (al contrario de las otras dos submuestras). Para la primera submuestra ningun factor es relevante, para la segunda solo el 3 y para la ultima submuestra son los 3 importantes. SK5 no es relevante en la submuestra 1 y 2 y SK4 no es relevante en la segunda muestra. Todo esto respecto al IMCE.